

Siddeshwar 'Sid' Raghavan

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RESEARCH INTERESTS

Continual Learning, Multi-Modal Foundational Models, 3D Vision, Casual Inference and Reasoning

EDUCATION

Purdue University

Ph.D. Electrical and Computer Engineering

West Lafayette, Indiana, USA

Jan 2022 – Present

University of Wisconsin-Madison

M.S Electrical Engineering (Research)

Madison, Wisconsin, USA

Aug. 2019 – May 2021

PSG College of Technology

B.E Electronics and Communication Engineering

Coimbatore, India

Jul. 2014 - May 2018

RESEARCH EXPERIENCE

Graduate Research Assistant - Advised by [Dr. Fengqing Zhu](#), [Dr. Edward J. Delp](#) May 2022 – Present

VIPER Lab, Purdue University

- Designed and developed an online **continual learning** computer vision algorithm to classify real-world food consumption patterns generated by our **probabilistic framework** (Skills: **Python, PyTorch, CUDA, Linux, Git**) [[WACV 2024](#)]
- Researched and developed an online continual learning algorithm for long-tailed (imbalanced) data using a **two-stage contrastive learner** framework and a **gradient balancing loss** to classify patterns, achieving an average 15% improvement over existing solutions [[CVPR 2024](#)]
- Researched and built an intelligent augmentation technique using vision language alignment for extreme imbalances and long-tailed scenarios for exemplar free continual learning [[AAAI 2026](#)]
- Developing **asynchronous audio-video** continual learning segmentation frameworks utilizing **causal inference** in egocentric videos [[On going](#)]
- Developing multi-modal hierarchical vision algorithms for 4D scene understanding [[On going](#)]

Graduate Research Assistant - Advised by [Dr. Yin Li](#)

Jan 2020 – May 2021

University of Wisconsin-Madison

- Developed a computer vision system for regressing and reconstructing intensity images from **Non-Line-Of-Sight(NLOS)** measurements using **3D multi-modal deep learning** models with a PSNR of 33.7 dB (Skills: **Python, PyTorch, Blender, Linux, Git, Docker**) [[IEEE TPAMI 2022](#)]
- Final thesis - **2D NLOS Human Pose Estimation** using a hybrid CNN and LSTM network.

TEACHING EXPERIENCE

Graduate Teaching Assistant – VIP Projects

Aug 2024 – Present

Purdue University

- Designed high-level project objectives**, weekly/monthly milestones, and a grading framework to guide student progress, **prioritizing their learning**.
- Mentored** undergraduate honors students through regular meetings, providing technical guidance and **measuring outcomes for project success**.
- Supported infrastructure setup** by configuring backend systems and hosting Gogs on lab servers to manage student code and deliverables.
- Instructors: **Carla Zoltowski, Edward Delp, Fengqing Zhu**

Graduate Teaching Assistant – Electrical Engineering Fundamentals II

May 2024 – Aug 2024

Purdue University

Lead Graduate Teaching Assistant – Digital Fundamentals

Jan 2021 – May 2021

University of Wisconsin–Madison

- Designed a semi-supervised grading framework in ModelSim to streamline evaluation, enabling TAs to devote more time to student support.
- Recognized with a Teaching Excellence Award during COVID for fostering student engagement and community in a virtual environment.
- Instructor: **Eric Hoffman**

INTERNSHIPS

Givens Research Associate Intern - Advised by **Dr. Tanwi Mallick**

May 2025 - July 2025

Argonne National Laboratory, USA

- Selected as **one of thirteen** Givens Associates this year out of 400 interns
- Developed a multi-agentic LLM framework in a post-training setup for scientific coding and reasoning. Surpassed current works by 10%. [\[NeurIPS 2025\]](#) [\[Paper\]](#)

Engineering Intern

Sept 2018– May 2019

Adori Labs, India

- Identified emerging markets for audio engineering and engagement and rapidly prototyped solutions for the domains. I developed a wake up voice SDK for the Adori player, Amazon Alexa and Google Home compatible frameworks and an in-house optimized search module with Elasticsearch. (**Skills: Python, Swift, C++**, AWS/GCloud, REST APIs, User-study, UX design)

Research Intern - advised by **Dr. Rajbabu Velmurugan**

Jun 2017 – Jul 2017

Indian Institute of Technology Bombay, India

- I worked on a weakly supervised **Visual SLAM** project and developed a single camera system (rather than a conventional multi-camera setup)

Research Intern - advised by **Dr. Ashok Jhunjunwala**

Jun 2016 – Jul 2016

Indian Institute of Technology Madras, India

- Designed and built a real-time battery state of health monitor for lithium ion cells with TI low power boards

RESEARCH PUBLICATIONS

1. PANDA - Patch And Distribution-Aware Augmentation for Long-Tailed Exemplar-Free Continual Learning (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) [\[AAAI 2026\]](#) [\[Paper\]](#)
2. MOSAIC: Multi-agent Orchestration for Task-Intelligent Scientific Coding (**Siddeshwar Raghavan**, Tanwi Mallick) [\[NeurIPS 2025\]](#) [\[Paper\]](#)
3. DELTA: Decoupling Long-Tailed Online Continual Learning (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) [\[CVPR Workshop 2024\]](#) [\[Paper\]](#) [\[Code\]](#)
4. Online Class-Incremental Learning for Real-World Food Image Classification (**Siddeshwar Raghavan**, Jiangpeng He, Fengqing Zhu) [\[WACV 2024\]](#) [\[Paper\]](#) [\[Code\]](#)
5. MetaFood3D: Large 3D Food Object Dataset with Nutrition Values (Yuhao Chen*, Jiangpeng He*, Gautham Vinod*, **Siddeshwar Raghavan***, Chris Czarnecki, Jinge Ma, Talha Ibn Mahmud, Dayou Mao, Saejith Nair, Pengcheng Xi, Alexander Wong, Edward Delp, Fengqing Zhu) [\[ArXiv 2024\]](#) [\[Paper\]](#) [\[Website\]](#) (* Equal contribution)
6. MFP3D: Monocular Food Portion Estimation Leveraging 3D Point Clouds (Jinge Ma, Xiaoyan Zhang, Gautham Vinod, **Siddeshwar Raghavan**, Jiangpeng He and Fengqing Zhu) [\[MADiMa 2024\]](#) [\[Paper\]](#)
7. Physics to the rescue: Deep non-line-of-sight reconstruction for high-speed imaging (Fangzhou Mu, Sicheng Mo, Jiayong Peng, Xiaochun Liu, Ji Hyun Nam, **Siddeshwar Raghavan**, Andreas Velten, Yin Li) [\[IEEE TPAMI 2022\]](#) [\[Paper\]](#) [\[Code\]](#)

PROFESSIONAL SERVICE AND LEADERSHIP ---

Professional Service

- **Workshop Organizer – MetaFood, CVPR 2025**
Main organizer of an international workshop featuring two challenges: (1) 3D food reconstruction from monocular images, and (2) vision–language models for interpreting textual food descriptions.
- **Reviewer – Premier Machine Learning and Computer Vision Venues**
Served as reviewer for top conferences and journals, contributing to the scholarly review process. [ICLR-26, AAAI-26, WACV-25, AAAI-25, NeurIPS-25, ICPR-24, MMSP-24, Elsevier, ICIP-24, CVPRW-24, ICME-23, CVPR-23, NeurIPS-23]
- **Judge – Undergraduate Research Conference, Purdue (2025)**
Lead graduate judge evaluating research posters and oral presentations across diverse disciplines.

Leadership and Outreach

- **Graduate Student Mentor and Teaching Leadership (VIP, Purdue University)**
Led project frameworks in signal processing and computer vision, mentored undergraduate honors students, and served as Head TA mentoring other teaching assistants.
- **Workshop Instructor – Deep Learning (2025, Purdue University)**
Conducted a 2-hour professional development workshop for 50+ undergraduate honors students, introducing fundamentals of deep learning.
- **Purdue Graduate Student Government – Life Team Co-Chair**
Managed funds and organized large-scale social and well-being events serving 10,000+ graduate students.
- **Teaching Excellence during COVID (UW–Madison)**
Recognized for cultivating virtual student communities and enhancing engagement as Head TA during remote instruction.

MENTORING EXPERIENCE ---

Undergraduate Students

- Joon Kim — Accepted into the M.S. ECE program, Purdue University
- Piotr Nabrzyski — Awarded Summer Research Fellowship, University of Houston
- Wei Lun Chu - Undergraduate researcher
- Raymond Chi - Undergraduate researcher

Graduate Students

- Jinge Ma — Co-authored paper accepted at MADiMa 2024
- Bruce Coburn - Junior PhD student in VIPER Lab